Fair Test Practical · Teacher and Cover Plan

Learning goal: I can record repeated results, spot an odd reading, and use the data to answer our question.

Scheme Outcome

TRACK Science scheme of work: Forces and Fair Testing

Carry out a fair-test force investigation with repeated measurements, identify any anomalous result, and draw a conclusion from the data.

Success Criteria

I can follow our fair-test plan and record three trials for each parachute size.

I can spot an anomalous (odd) reading that does not fit the pattern.

I can write a conclusion that quotes the numbers and links canopy size to air resistance.

Model Sequence (I do)

First, check the three repeats for each size: Small (1.2, 1.3, 1.1 s), Medium (2.1, 2.0, 2.2 s), Large (3.4, 3.5, 3.3 s).

Second, find the typical (middle or mean) value for each: 1.2 s, 2.1 s, and 3.4 s.

Third, link the numbers to the science: larger canopy area increases air resistance, slowing the fall.

When writing a conclusion, always quote at least two numbers with their units (e.g. 3.4 s vs 1.2 s).

Next, explain the mechanism: a larger canopy pushes against more air, creating greater air resistance.

Finally, suggest one improvement, such as using a taller drop or video timer to reduce stopwatch reaction error.

We Do & Hinge Check

For the 30x30 cm parachute, a group recorded: Trial 1 = 3.4 s, Trial 2 = 1.3 s (canopy did not open), Trial 3 = 3.5 s, Retest = 3.3 s. Which value is the anomaly?

Answer: A. 1.3 s is the anomalous reading caused by the unopened canopy; the valid trials cluster around 3.4 s.

Safe Practical & Card Setup

Carry out the 3-size parachute drops (or use the supplied dataset) and complete the results table and conclusion.

Kit: Three square canopies (10x10, 20x20, 30x30 cm), 10 g mass, metre rule (1.50 m mark), and stopwatch.

Release from 1.50 m while standing flat on the floor. Keep the landing zone clear.

You Do Routes (Supported, Independent, Aspire)

Supported: Record the typical times (1.2 s, 2.1 s, 3.4 s), circle which parachute fell slowest, and complete the sentence. (Help: 3.4 s is the longest time. The __ canopy fell slowest because it had the most air resistance.)

Independent: Complete the typical fall times for Small

(1.2 s), Medium (2.1 s), and Large (3.4 s), identify the anomaly in the sample row, and write a conclusion quoting the numbers. (Help: As canopy size increased from 10x10 cm to 30x30 cm, fall time increased from __ s to __ s because a larger canopy creates more __.)

Aspire: Explain why excluding the flagged anomaly (1.3 s) gives a more accurate mean for the 30x30 cm canopy, and evaluate stopwatch reaction-time error. (Help: Including the tangled 1.3 s drop would pull the mean down unfairly. Stopwatch reaction time (~0.2 s) matters most on the __ fall times.)